

Predicting Daily Household Energy Consumption Using Kernel Methods and Gaussian Process Regression

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Abstract—Accurate forecasting of residential energy consumption is essential for demand planning, grid stability, and energy-efficient resource allocation. This study investigates a suite of kernel-based learning methods, including Kernel Ridge Regression (KRR) and Gaussian Process Regression (GPR), applied to the UCI Household Electric Power Consumption dataset. The objective is to compare univariate and multivariate prediction models, evaluate nonlinear kernels, and analyze their performance relative to baseline forecasting methods. Daily energy consumption values were derived through resampling and used to construct both lag-based univariate models and multivariate models incorporating reactive power, voltage, intensity, and sub-metering features. Results show that multivariate kernel models significantly outperform univariate models and baselines, with Multivariate KRR (Polynomial kernel) achieving the best performance (RMSE = 0.2508, MAE = 0.1856). The study highlights the interpretability of GPR and the expressive power of kernel methods for modeling nonlinear time-series dynamics.

I. INTRODUCTION

Forecasting household energy consumption has become increasingly important with the growth of smart grids, dynamic pricing, and renewable load balancing. Traditional linear models struggle to capture nonlinear interactions and seasonal effects present in electrical consumption data. Kernel-based learning methods, specifically Kernel Ridge Regression (KRR) and Gaussian Process Regression (GPR), offer flexible nonlinear modeling capabilities through reproducing kernel Hilbert space (RKHS) representations.

This paper develops and compares several kernel methods on a daily energy consumption forecasting task. Both univariate time-series models (using lagged consumption values) and multivariate models (including electrical and sub-metering covariates) are evaluated. Our goals are:

- 1) To quantify the benefits of multivariate modeling over univariate modeling;
- 2) To compare the performance of RBF, Polynomial, Periodic, Matérn, and composite kernels;
- 3) To analyze GPR hyperparameters and posterior variance for interpretability.

II. DATASET AND PREPROCESSING

We use the UCI Household Electric Power Consumption dataset, containing minute-level measurements from December 2006 to November 2010. The raw data includes global active and reactive power, voltage, intensity, and three sub-metering variables. The dataset contains missing values represented by

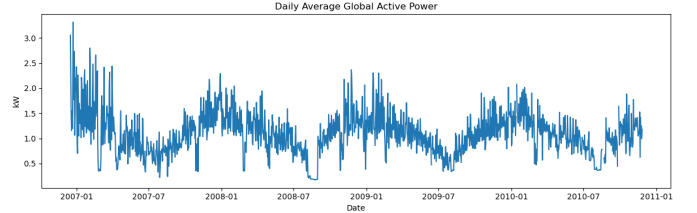


Fig. 1: Daily average global active power over the full observation period. The series exhibits smooth long-term variation and moderate seasonality.

“?”. All values were converted to floats, missing entries were imputed via forward and backward filling, and a combined timestamp index was constructed.

To reduce noise and align with long-term consumption trends, the data was resampled into daily averages. Let y_t denote the daily global active power (kW). For univariate models, features consisted of:

$$X_t = \{y_{t-1}, y_{t-7}, y_{t-30}\}.$$

For multivariate models, we construct a 6-dimensional feature vector using lagged electrical measurements. Specifically, the input at time t is defined as

$$Z_t = \begin{bmatrix} \text{ReactivePower}_{t-1} \\ \text{Voltage}_{t-1} \\ \text{Intensity}_{t-1} \\ \text{Sub1}_{t-1} \\ \text{Sub2}_{t-1} \\ \text{Sub3}_{t-1} \end{bmatrix} \in \mathbb{R}^6.$$

This vector captures the electrical characteristics of the household on the previous day and provides additional context beyond lagged active power alone. These features were selected because they exhibit meaningful correlations with daily energy consumption and are physically interpretable components of the household’s electrical load.

III. METHODOLOGY

A. Kernel Ridge Regression

Kernel Ridge Regression (KRR) combines ridge regression with kernel methods in an RKHS. It solves:

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \left(\sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{H}}^2 \right), \quad (1)$$

where $\mathcal{H}$ is a reproducing kernel Hilbert space with kernel $k(\cdot, \cdot)$, and $\lambda > 0$ is a regularization parameter.

By the representer theorem, the solution can be expressed as:

$$\hat{f}(x) = \sum_{i=1}^n \alpha_i k(x, x_i), \quad (2)$$

with dual coefficients

$$\alpha = (K + \lambda I)^{-1} \mathbf{y}, \quad (3)$$

where $K \in \mathbb{R}^{n \times n}$ is the Gram matrix with $K_{ij} = k(x_i, x_j)$.

We evaluate the following kernels:

- **RBF kernel:**

$$k_{\text{RBF}}(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right), \quad (4)$$

where ℓ is the length-scale.

- **Polynomial kernel:**

$$k_{\text{Poly}}(x, x') = (x^\top x' + c)^d, \quad (5)$$

where $c \geq 0$ is a bias term and d is the polynomial degree.

B. Gaussian Process Regression

A Gaussian Process (GP) is a collection of random variables, any finite number of which have a joint Gaussian distribution:

$$f(x) \sim \mathcal{GP}(m(x), k(x, x')), \quad (6)$$

where $m(x)$ is the mean function (taken as zero) and $k(x, x')$ is the covariance kernel.

Given training data $(X, \mathbf{y})$ with $X = [x_1, \dots, x_n]^\top$ and noisy observations $y_i = f(x_i) + \epsilon_i$, with $\epsilon_i \sim \mathcal{N}(0, \sigma_n^2)$, the joint prior over training outputs $\mathbf{y}$ and a test point $f_* = f(x_*)$ is:

$$\begin{bmatrix} \mathbf{y} \\ f_* \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} K + \sigma_n^2 I & k_* \\ k_*^\top & k(x_*, x_*) \end{bmatrix}\right), \quad (7)$$

where $K_{ij} = k(x_i, x_j)$ and $k_* = [k(x_*, x_1), \dots, k(x_*, x_n)]^\top$.

The posterior predictive distribution is Gaussian with mean and variance:

$$\hat{y}_* = \mathbb{E}[f_* | \mathbf{y}] = k_*^\top (K + \sigma_n^2 I)^{-1} \mathbf{y}, \quad (8)$$

$$\text{Var}(y_*) = k(x_*, x_*) - k_*^\top (K + \sigma_n^2 I)^{-1} k_*. \quad (9)$$

We consider the following kernels for GPR:

- **RBF kernel** as in (5);
- **Periodic kernel** (ExpSineSquared) for seasonality:

$$k_{\text{Per}}(x, x') = \exp\left(-\frac{2 \sin^2(\pi \|x - x'\|/p)}{\ell^2}\right), \quad (10)$$

where p is the period;

- **Additive RBF + Periodic kernel:**

$$k_{\text{RBF+Per}}(x, x') = k_{\text{RBF}}(x, x') + k_{\text{Per}}(x, x'); \quad (11)$$

- **Matérn kernel** with $\nu = 1.5$:

$$k_{\text{Mat}}(r) = \sigma^2 \left(1 + \frac{\sqrt{3}r}{\ell}\right) \exp\left(-\frac{\sqrt{3}r}{\ell}\right), \quad (12)$$

where $r = \|x - x'\|$.

C. Hyperparameter Selection

For KRR, we tune the regularization parameter λ and kernel hyperparameters using grid search with a time-series aware cross-validation scheme (TimeSeriesSplit). The RBF kernel is parameterized by $\gamma = 1/(2\ell^2)$; typical grids include:

$$\lambda \in \{0.01, 0.1, 1.0, 10.0\}, \quad (13)$$

$$\gamma \in \{0.01, 0.1, 1.0, 10.0\}. \quad (14)$$

For the Polynomial kernel, we search over degree d and bias c :

$$d \in \{2, 3, 4\}, \quad (15)$$

$$c \in \{0.0, 1.0, 5.0\}. \quad (16)$$

For GPR, kernel hyperparameters such as length-scale ℓ , signal variance σ^2 , noise variance σ_n^2 , and periodicity p are optimized by maximizing the log-marginal likelihood:

$$\log p(\mathbf{y} | X, \theta) = -\frac{1}{2} \mathbf{y}^\top K_\theta^{-1} \mathbf{y} - \frac{1}{2} \log |K_\theta| - \frac{n}{2} \log 2\pi, \quad (17)$$

where θ denotes the kernel parameters. This provides a principled Bayesian approach to model complexity control and automatically balances data fit with smoothness.

IV. EXPERIMENTS

All features were standardized to zero mean and unit variance using statistics estimated from the training set. Univariate models used lagged daily consumption values as inputs, while multivariate models combined lagged consumption with lagged electrical and sub-metering covariates.

Baselines included:

- **Naive:** $\hat{y}_t = y_{t-1}$;
- **MA(7):** constant forecast equal to the final 7-day rolling mean of the training period.

KRR models with RBF and Polynomial kernels were trained for both univariate and multivariate feature sets. GPR models were evaluated with RBF, Periodic, RBF+Periodic, and Matérn kernels. TimeSeriesSplit with 5 folds was used for KRR hyperparameter selection. GPR hyperparameters were initialized with reasonable scales and optimized via gradient-based marginal likelihood maximization.

V. RESULTS

Table I summarizes all model results on the 2010 test set. Performance is measured using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

Figure 2 illustrates example predictions on the test set for the best univariate and multivariate models, while Fig. 3 provides a visual comparison of RMSE across all models.

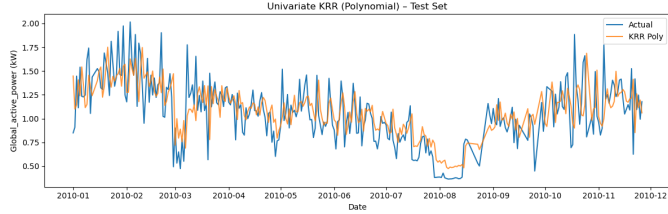
VI. DISCUSSION

A. Multivariate Models Outperform Univariate Models

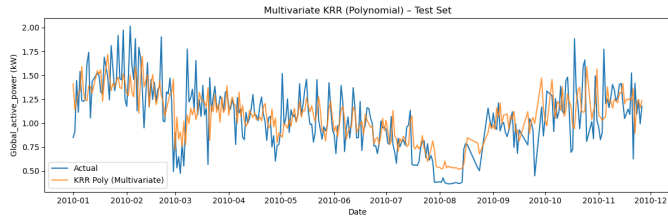
All multivariate kernels outperform their univariate counterparts. This confirms that reactive power, voltage, intensity, and sub-metering variables contain predictive information not captured by lagged consumption alone. The gap between the best multivariate and best univariate models is non-trivial, demonstrating the benefit of leveraging additional covariates.

TABLE I: Model Performance Comparison (Sorted by RMSE)

Model	RMSE	MAE
MULTI — KRR Poly	0.2508	0.1856
MULTI — KRR RBF	0.2521	0.1902
MULTI — GPR RBF	0.2524	0.1860
MULTI — GPR Matérn	0.2526	0.1874
UNI — GPR RBF	0.2539	0.1902
UNI — GPR RBF+Periodic	0.2550	0.1910
UNI — KRR Poly	0.2561	0.1926
UNI — KRR RBF	0.2566	0.1954
UNI — GPR Periodic	0.2588	0.1970
UNI — Naive	0.3032	0.2200
UNI — MA(7)	0.5710	0.4907



(a) Univariate model: example predictions vs. actual daily consumption.



(b) Multivariate KRR (Polynomial): predictions vs. actuals on the test set.

Fig. 2: Predicted vs. actual daily global active power for representative models.

B. Effect of Kernel Choice

Multivariate KRR with a Polynomial kernel achieves the lowest RMSE, suggesting that nonlinear feature interactions across covariates are important. While the RBF kernel captures smooth trends in the feature space, the Polynomial kernel effectively models higher-order relationships, which appear to matter for this task.

RBF-based GPR models perform almost identically to KRR with RBF kernels. This is consistent with the close relationship between KRR and GPR: both impose a quadratic penalty in the RKHS, but GPR additionally provides a full Bayesian treatment with uncertainty estimates. The Matérn kernel performs similarly to RBF, indicating that the daily series is reasonably smooth but still exhibits some roughness that Matérn can accommodate.

Purely Periodic kernels underperform RBF-based kernels, particularly in the univariate setting. Daily averaging tends to smooth out sharp seasonal effects and introduces slow non-stationary trends that are not purely periodic. As a result, the RBF and Polynomial kernels, which can model more general

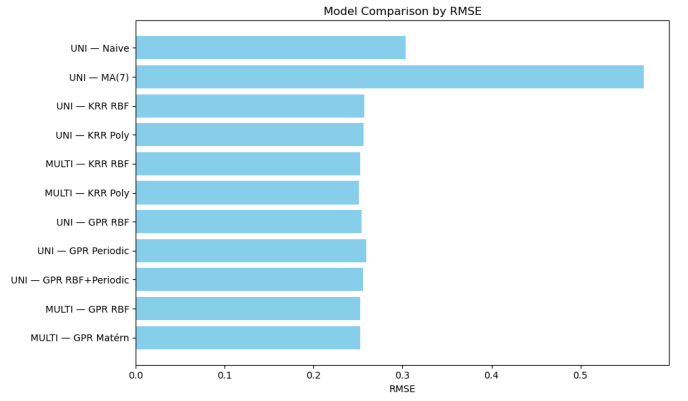


Fig. 3: RMSE comparison for all evaluated models. Multivariate kernel methods clearly outperform univariate models and baselines.

smooth structure, are better suited to this data.

C. Hyperparameter Behavior and Interpretation

Hyperparameter tuning plays a central role in the performance of kernel methods. In KRR, the regularization parameter λ controls the trade-off between data fit and model complexity: small values of λ allow the model to fit the training data more closely, while larger values enforce smoother solutions in the RKHS. The RBF kernel parameter γ (or equivalently the length-scale ℓ) determines how quickly correlations decay in feature space; moderate values are typically preferred for smooth but not overly rigid fits.

For Polynomial kernels, the degree d controls the complexity of the feature mapping. Low degrees behave similarly to linear or mildly nonlinear models, whereas higher degrees risk overfitting if not regularized properly. The bias term c controls how strongly lower-order interactions contribute relative to higher-order terms.

In GPR, the length-scale ℓ in RBF and Matérn kernels controls how rapidly the function can vary with respect to the inputs. Larger length-scales correspond to smoother functions, while smaller length-scales allow rapid variation but can lead to overfitting. The noise variance σ_n^2 captures observation noise and unresolved variability; its inferred magnitude provides insight into how much of the variation is explained by the kernel versus treated as noise. Periodicity parameters p in Periodic kernels encode the assumed seasonal cycle; mismatched or overly rigid periodicity can degrade performance when the data exhibit evolving or imperfect cycles.

Overall, the strong performance of RBF and Polynomial kernels with moderate regularization supports the view that daily household energy consumption can be modeled as a relatively smooth function of past consumption and electrical covariates, with nonlinear interactions playing an important role.

VII. CONCLUSION

Kernel-based learning methods provide strong predictive performance for daily household energy consumption. Mul-

tivariate models significantly improve accuracy relative to univariate lag-based models, and Polynomial kernels in particular capture essential nonlinear interactions among covariates. Gaussian Process models offer comparable accuracy to KRR with the added benefit of interpretable uncertainty estimates through posterior variance.

Future work may include multi-output GPR models, deep kernel learning, or the use of locally periodic kernels to better capture slowly evolving seasonal patterns. Additionally, incorporating exogenous variables such as weather or pricing information could further enhance forecasting performance.

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